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Fraud Detection using Graphical Neural Networks

Abstract:

Traditional models treat transactions independently, missing the hidden relationships between transactions. In this project, we convert tabular transaction data into a graph structure where nodes represent transactions and edges capture shared attributes like card, device, or email. We then apply Graph Neural Networks (GraphSAGE and GAT) to leverage these connections for fraud detection. Our results show that graph-based models outperform the baseline MLP, proving that relational information significantly improves fraud detection.

Dataset:

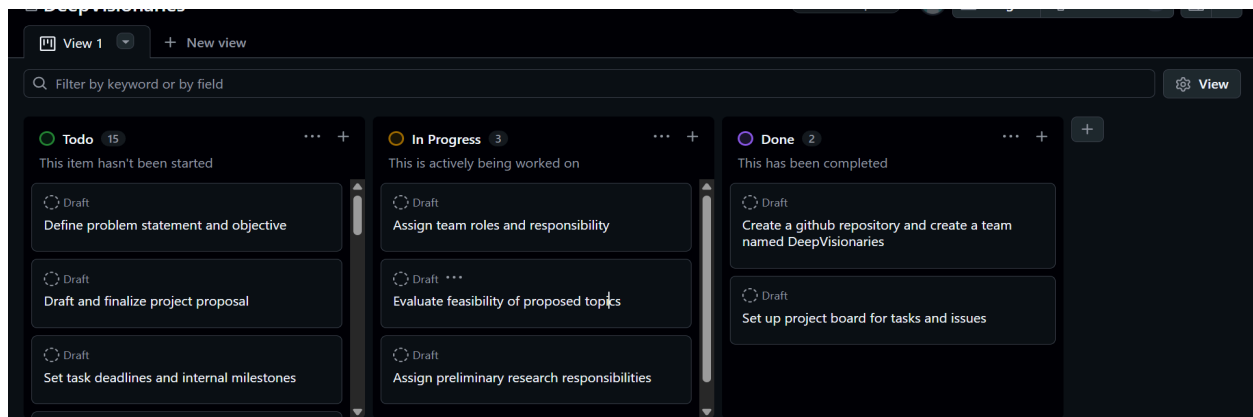
We used the IEEE-CIS Fraud Detection dataset (Kaggle), which contains over 590,000 transactions with both transaction details and identity information. We merged *train_transaction* and *train_identity* using *TransactionID*. The dataset is highly imbalanced (~3.5% fraud). Basic preprocessing included log transformation of transaction amount, handling missing values, and feature scaling.

Evaluation:

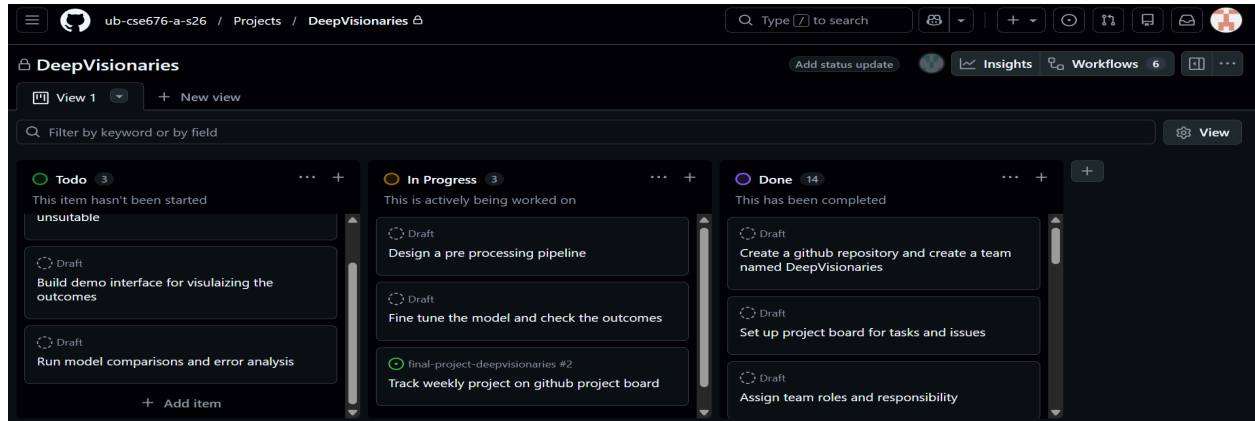
We evaluated models using ROC-AUC and F1-score, since the dataset is highly imbalanced. AUC measures overall ranking performance, while F1 helps balance precision and recall for fraud detection. Thresholds were tuned using validation data to maximize F1-score. Graph-based models (GraphSAGE, GAT) were compared against a baseline MLP.

Project Milestones (PM Tool):

1. INITIAL PLANNING FOR THE PROJECT



2. TARGET ACHIEVED TILL CHECKPOINT



3. PLANNED WORK FOR FINAL SUBMISSION

